

SUNREY

TECHNICAL WHITE PAPER

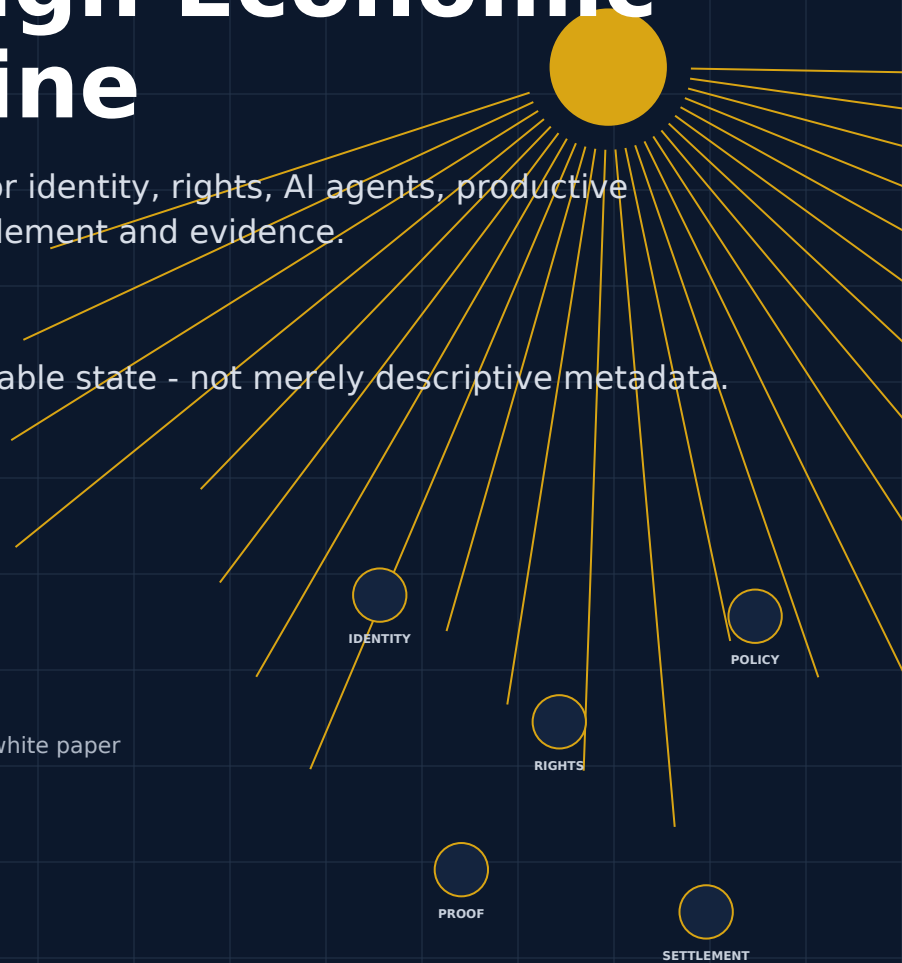
SunRey Blockchain The Sovereign Economic State Machine

A purpose-built economic protocol for identity, rights, AI agents, productive capacity, policy-aware markets, settlement and evidence.

Protocol thesis

Economic context should be enforceable state - not merely descriptive metadata.

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0. Abstract

SunRey Blockchain is designed as a sovereign economic state machine for a post-AI economy. Its objective is not to reproduce a generic smart-contract network with a new token. The architecture treats economically meaningful actors, rights, permissions, provenance, productive capacity, jurisdiction, policy, delivery and evidence as first-class state alongside native assets and settlement.

The protocol is intended to support two complementary economic domains on one sovereign base layer. SunRey Coin coordinates the human-information and participation economy; MoonRey Coin coordinates autonomous productive capacity and machine-generated output. Both are native network assets rather than ERC-20 tokens dependent on another blockchain for core settlement.

SunRey Exchange extends the model into a Universal Economic Exchange. Fungible digital assets can use conventional market structures, while information rights, secure compute and productive capacity can use different mechanisms such as cohort requests, compute jobs, RFQs, auctions and capacity reservations. Matching can evaluate not only price and quantity but identity, purpose, consent, jurisdiction, provenance, asset state, delivery conditions and policy.

CORE ARCHITECTURE PRINCIPLE

SunRey Blockchain establishes identity, rights, policy state, provenance and settlement. Sensitive source data stays protected. AI may discover and propose. Deterministic authorization and risk controls determine what may execute. The chain records authoritative economic state and evidence.

Design boundary. The differentiated thesis is the integration and first-class treatment of economic context. Consensus algorithms, virtual machines, zero-knowledge systems, matching engines, oracles and other underlying techniques may use established methods. Formal novelty and patentability require separate prior-art and claims analysis.

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READING FRAME

This paper describes the protocol architecture and economic design. It does not represent a mainnet launch, license, custody approval, asset-backing arrangement, or authorization to operate regulated markets.

1. From Blockchain to Economic State Machine

SunRey begins with the blockchain primitive of deterministic state transition, then expands the definition of state to include the context that gives an economic event meaning.

Most blockchains are optimized around a narrow set of native concepts: accounts or objects, balances, signatures, program execution and consensus on state transitions. Everything else is usually delegated to arbitrary smart contracts or off-chain application logic. That model is powerful, but it does not automatically make identity, legal rights, consent, machine authority, productive capacity or delivery conditions legible to the protocol.

SunRey proposes a richer economic ontology. A high-context transaction should be able to answer who or what is acting, under whose authority, which economic object is involved, which right is being transferred or exercised, what purpose is authorized, which jurisdiction and legal entity govern, what external fact or productive output has been verified, what model or agent proposed the action, what policy allowed it, what consideration was paid and what evidence proves completion.

ANATOMY OF A HIGH-CONTEXT ECONOMIC EVENT

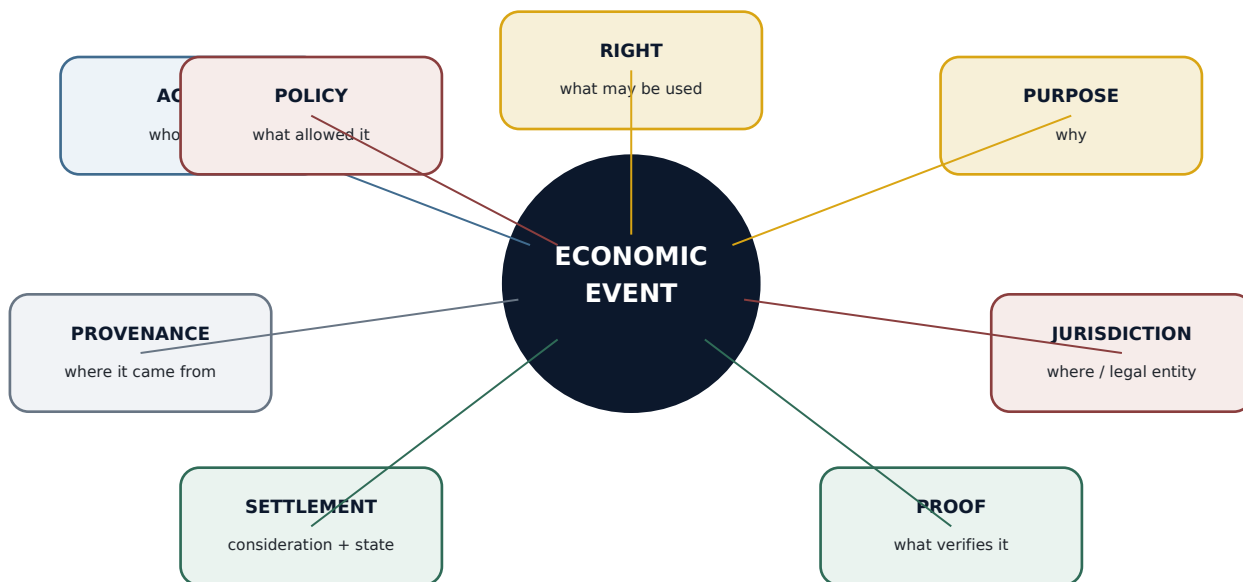


Figure 1. The protocol treats economic context as state that can be evaluated before execution and reconstructed after settlement.

THE DESIGN SHIFT

A generic token transfer proves that value moved. A sovereign economic state machine is designed to prove what the transaction meant, why it was allowed and what evidence made settlement valid.

2. What the Chain Treats as Native Economic State

The state machine is organized around a controlled set of recurring economic primitives rather than an unlimited promise that every real-world concept belongs directly on-chain.

Native state	Economic meaning
Actors and identities	Humans, pseudonymous humans, enterprises, AI agents, robots, devices and productive assets.
Ownership and non-ownership rights	Access, compute, usage, lease, delivery, participation and revocation rights.
Consent and purpose	Scope, duration, compensation, transferability, permitted models and revocation state.
Provenance	Origin, verification source, transformation lineage and model/agent references.
Productive state	Capacity, availability, utilization, delivered output and settlement condition.
Jurisdiction and policy	Legal-entity mapping, product capability, eligibility and transaction restrictions.
Attestation and proof	Credentials, oracle proofs, evidence references and delivery claims.
Native assets	SunRey Coin and MoonRey Coin with distinct economic roles and governance.

The purpose of this ontology is not to turn every fact into a token. It is to give economically significant relationships a consistent representation. Ownership can be distinct from usage. A data subject can grant a time-limited compute right without transferring the underlying record. A productive facility can advertise verified capacity without implying tokenized legal ownership of the facility itself.

PROTOCOL RULE

Possession does not automatically imply unrestricted use. Ownership state and rights state can be distinct, and execution must respect both.

3. The Sovereign Economic State Machine

SUNREY SOVEREIGN ECONOMIC STATE MACHINE

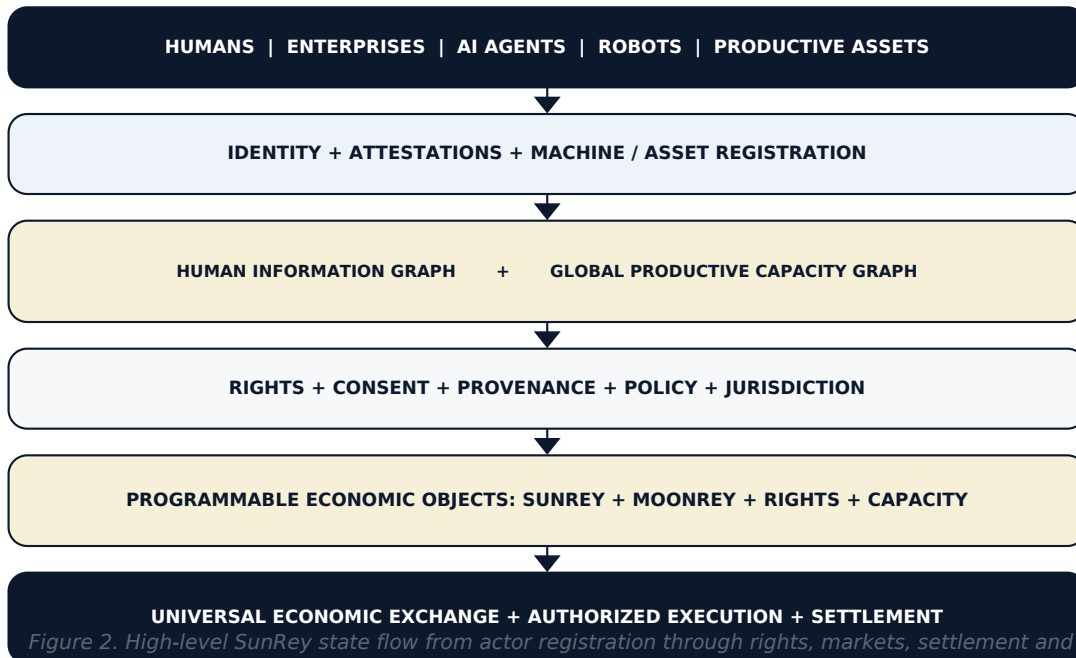


Figure 2. High-level SunRey state flow from actor registration through rights, markets, settlement and proof.

The state machine places identity and attestation above economic objects because an asset or right is meaningful only in relation to the actor authorized to control or exercise it. The Human Information Graph and Global Productive Capacity Graph provide permission-aware context for the two sides of the economy. Rights, consent, provenance, policy and jurisdiction determine which interactions are permissible.

The programmable economic object layer then represents native assets and typed rights. SunRey Exchange provides discovery and matching; authorized execution and settlement resolve the transaction; and the blockchain finalizes the authoritative state with proof and evidence references.

SOVEREIGNTY PRINCIPLE

External chains may interoperate later, but they remain external networks. SunRey Blockchain is intended to remain the system of record for native SunRey and MoonRey economic activity.

4. The Dual-Economy Model: SunRey + MoonRey

The protocol separates the human economic layer from the autonomous productive layer while allowing both to interact through one trust, policy and settlement system.

TWO ECONOMIC DOMAINS, ONE SOVEREIGN TRUST LAYER

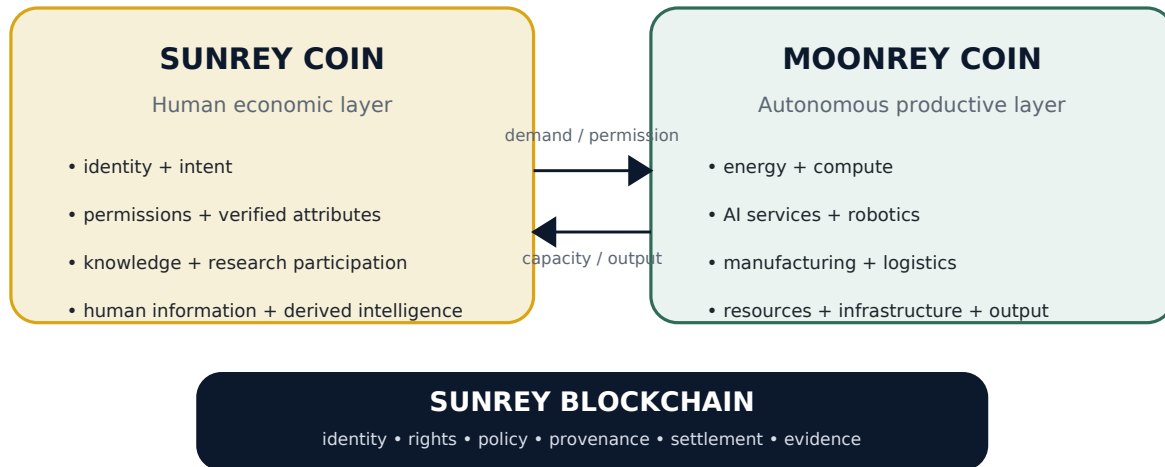


Figure 3. Human demand and permission interact with autonomous capacity and output on a common sovereign base layer.

SunRey represents economic activity associated with people, communities and permissioned human information: identity, intent, verified attributes, knowledge, research participation, preferences, reputation and derived intelligence. MoonRey represents the productive systems that supply goods and services: energy, compute, AI services, robotics, manufacturing, logistics, real estate, infrastructure, resources and machine-generated output.

The two sides are complementary. Humans express goals, demand, preferences and permissions. Automated systems allocate productive capacity to satisfy those demands. A shared protocol can make the relationship programmable while keeping the two native assets economically distinct.

ECONOMIC INTERPRETATION

SunRey Coin and MoonRey Coin are best described as native coordination and settlement assets whose utility is generated by network activity. Any future claim that an asset is legally backed by people, data, energy, real estate or other resources would require separate reserve, custody, valuation, redemption and legal structures.

5. SunRey Coin: The Human Economic Layer

SunRey Coin is the native economic coordination asset for the Human Information Network and the broader human-participation economy. The design does not treat a person or a person's raw data as collateral. It treats authorized participation, rights and verified economic activity as the relevant economic layer.

Human-economy primitive	Representative activity
Identity and verified credentials	Attested identity state, professional or educational credentials and eligibility proofs.
Intent and preferences	Goals, purchase intent, consumer demand and personal mandates where authorized.
Permissioned information	Financial, behavioral, mobility, lifestyle, health or research attributes only where lawful and explicitly authorized.
Participation and contribution	Research, community, economic activity, accepted recommendations and other verified contributions.
Rights and consent	Purpose, scope, duration, revocation and compensation terms.
Derived intelligence	Authorized cohorts, eligibility proofs, attestations and compute-to-data outputs.

The protocol therefore separates identity from monetization. A person can be the subject of an attribute without the raw attribute becoming a tradable public object. The economic object can instead be a permission, an attestation, a compute right, a research participation contract or a compensation claim.

HUMAN INFORMATION PRINCIPLE
Value the authorized contribution or right - not the intrinsic worth of the person. Sensitive source data remains protected while permission, provenance, proof and settlement become machine-manageable.

6. MoonRey Coin: The Autonomous Productive Layer

MoonRey Coin is the complementary native asset for an economy in which AI systems, robots, infrastructure and resources perform an increasing share of productive work. Its central concept is productive capacity: the measurable ability of a system to provide economically useful output.

The architecture therefore begins with a different question from conventional asset tokenization: what useful output can this system reliably produce - where, when, at what cost, under whose control and with what reliability?

Productive pool	Representative capacity
Energy	generation, transmission, storage, charging and verified energy provenance
Compute	GPU-hours, CPU-hours, storage, bandwidth, inference and training capacity
AI services	autonomous research, software, analytics, planning and service output
Robotics	robot-hours, fleet utilization, delivery, warehouse or manufacturing capacity
Manufacturing	machine-hours, factory throughput, production runs and delivered goods
Food / water	agricultural output, processing, cold chain, treatment and desalination capacity
Real estate / lodging	room-nights, occupancy, warehouses, data centers and usable productive space
Mobility / logistics	vehicle-days, rides, freight, transport and delivery capacity
Infrastructure / resources	telecom, grids, storage, minerals and legally recognized usage rights

MOONREY OBJECTIVE

Represent verifiable productive state and rights without pretending that every network object is automatically a legal claim on the underlying physical asset.

7. Protocol Component Architecture

SunRey Blockchain is intended as the sovereign base layer for the ecosystem. Its architecture combines conventional chain responsibilities with registries and control planes required for human information, AI agents, machine commerce and productive capacity.

Protocol component	Responsibility
Consensus & Validator Layer	Finality, validator accountability, network security and protocol governance.
Native Execution Runtime	Deterministic programmable execution for rights, markets, policy-aware objects and machine commerce.
Dual Native Asset Layer	Protocol-native SunRey Coin and MoonRey Coin with separately governed rules.
Identity & Actor Registry	Human, enterprise, AI-agent, robot, device and productive-asset identities.
Human Information Graph	Permission-aware human economic context, verified attributes, consent and contributions.
Global Productive Capacity Graph	Machine-readable energy, compute, factories, logistics, infrastructure and output.
Rights & Consent Registry	Ownership, access, compute, lease, usage, delivery and participation rights.
Provenance & Attestation Layer	Source verification, transformation lineage, oracle facts and proofs.
Policy & Jurisdiction Layer	Product capability, legal-entity mapping, eligibility and transaction restrictions.
AI / Robot Capability Layer	Capabilities, mandates, budgets, risk limits and revocation controls.
Oracle Network	Authenticated observations for production, capacity, price, asset state and delivery.
Settlement & Evidence Layer	Final state, receipts, evidence hashes and proof references.
Interoperability Gateway	Controlled adapters to external chains and financial rails.

The architecture deliberately avoids making any one external chain, provider, oracle or application the owner of SunRey-native truth. External systems contribute evidence or execution within defined interfaces; the canonical protocol owns native state.

8. Consensus, Finality and the Native Execution Runtime

The consensus layer should prioritize safety, deterministic finality, validator accountability, resilience and disciplined upgrades over headline claims about raw throughput. The specific consensus mechanism remains a formal architecture decision and should be selected through explicit trade-off analysis rather than branding.

Design objective	Protocol implication
Deterministic finality	Economic state should have a clear point at which a transaction becomes authoritative.
Validator accountability	Operator identity, policy, fault evidence and any slashing/accountability model should be explicit.
Resilience	The network should tolerate infrastructure failures and support recovery without ambiguous economic history.
Regional / institutional participation	Future sovereignty or regulatory requirements may require controlled validator topologies.
Upgrade discipline	Critical changes should use versioned activation, review, test networks and rollback/pause strategies where feasible.

The native execution environment should likewise favor predictable resource accounting, deterministic behavior, strong type safety or constrained contract interfaces, formal versioning and mature security tooling. The source architecture identifies a WASM-style runtime as a plausible direction, not a locked decision.

Sovereignty does not require reinventing every cryptographic primitive. The protocol can use established execution and security techniques while preserving sovereign control over the state model, native economic objects, governance and settlement rules.

ENGINEERING BOUNDARY

Consensus and runtime choices remain open architecture decisions. The white paper defines the economic ontology and authority model; it does not claim a final production consensus implementation.

9. Identity and Actor Primitives

ECONOMIC ACTORS ARE MORE THAN WALLETS

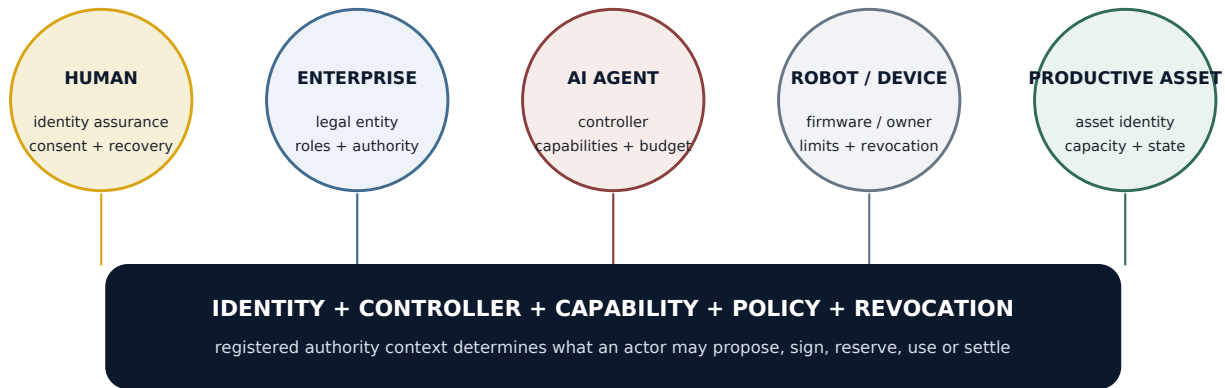


Figure 4. AI agents, robots and productive assets receive explicit identity and authority context rather than being treated as anonymous wallets.

AI agents and robots should not be represented merely as private keys controlling balances. They should have registered identities, owner/controller relationships, capability manifests, permitted actions, budget limits, jurisdiction and policy state, model or firmware references and revocation status.

The same principle applies to productive assets. A solar facility, data center, autonomous fleet or factory line can have an asset identity linked to ownership or control state, capacity, availability, quality attributes, oracle feeds and delivery history. The protocol can therefore evaluate both who controls the actor and what the actor is permitted to do.

MACHINE AUTHORITY RULE

An AI agent may discover and propose a purchase. A robot may hold a bounded operating capability. Neither should inherit unrestricted economic authority simply because it controls a wallet key.

10. Rights, Consent, Provenance, Policy and Jurisdiction

SunRey's strongest protocol distinction is the idea that economic rights and policy conditions should be represented close to the state they govern.

State domain	Questions the protocol should be able to answer
Rights	Is this ownership, access, compute, lease, usage, delivery or participation? Is transfer allowed?
Consent	Who authorized the use? For which scope and purpose? For how long? Can it be revoked?
Provenance	Where did the information or output originate? How was it transformed? Which model or machine produced it?
Policy	Which rule set evaluated the action? Which version was active? What condition allowed or blocked it?
Jurisdiction	Which legal entity and operating capability apply to this actor, object and market?
Evidence	What proof, attestation, oracle fact, receipt or delivery event supports the final state?

This model makes rights portable without requiring sensitive source data to be public. A user can grant a narrow research permission. An enterprise can buy a compute right. A buyer can reserve a quantity of productive capacity. Each can have a different expiration, transferability rule, jurisdiction, settlement condition and evidence requirement.

STATE BEFORE SETTLEMENT

Identity tells the protocol who or what is acting. Rights define what can be exercised. Consent and purpose define why. Policy and jurisdiction determine whether. Provenance and evidence explain what happened.

11. Native Economic Objects

The Exchange and surrounding applications require a canonical object standard that can represent more than fungible balances while preserving type-specific validation and policy.

Object type	Purpose	Illustrative fields
Fungible Asset	SunRey, MoonRey or approved digital instruments.	amount; issuer/network; owner; custody state; transfer rules
Verified Attribute	Claim about a person, machine or asset without unnecessary source disclosure.	issuer; subject; claim; validity; provenance; privacy level
Permission / Data Right	Purpose- and time-limited operation against protected information.	subject; requester; purpose; scope; duration; compensation; revocation
Compute Right	Right to run an approved model or query.	workload; data scope; environment; resource limit; output rule; price
Productive Capacity Right	Right to consume or reserve verified productive capacity.	asset ID; unit; capacity; window; location; quality
Output / Delivery Claim	Proof that a machine, facility or service delivered specified output.	producer; quantity; quality; time; oracle proof; settlement state
Attestation / Credential	Verifiable statement for identity, quality, eligibility or compliance.	issuer; subject; schema; validity; revocation; proof
Composite Economic Workflow	Bundle of interdependent rights and services.	component objects; dependencies; policy; settlement; evidence

A common object interface allows the protocol and Exchange to reason consistently about ownership, permission, capacity and delivery while still applying different execution and settlement rules to each class.

12. The Rich Transaction Envelope

RICH TRANSACTION ENVELOPE

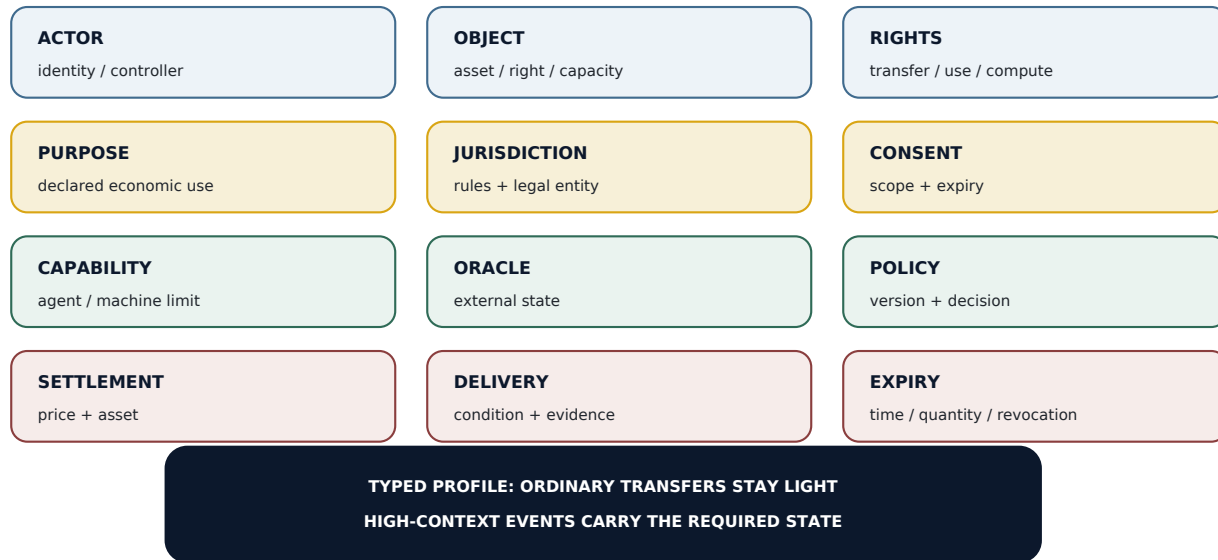


Figure 5. Typed transaction profiles allow high-context economic events to carry the conditions required for safe execution.

A SunRey transaction envelope can include actor identity, object type, rights being transferred or exercised, purpose, jurisdiction, consent reference, capability reference, oracle state, policy version, price, settlement asset, delivery condition, evidence requirements and expiration.

The design does not require every field on every transfer. Typed profiles can keep ordinary fungible transfers efficient while attaching richer state only when the economic object requires it. A compute-to-data job, for example, may need purpose, consent, workload limits, output rules and proof. A productive-capacity reservation may need asset identity, availability window, location, quality and delivery conditions.

TRANSACTION SEMANTICS

The transaction envelope is intended to make execution context verifiable rather than leaving critical meaning in an off-chain terms document that the protocol cannot evaluate.

13. SunRey Exchange: The Universal Economic Exchange

SUNREY EXCHANGE - UNIVERSAL ECONOMIC MARKET LAYER

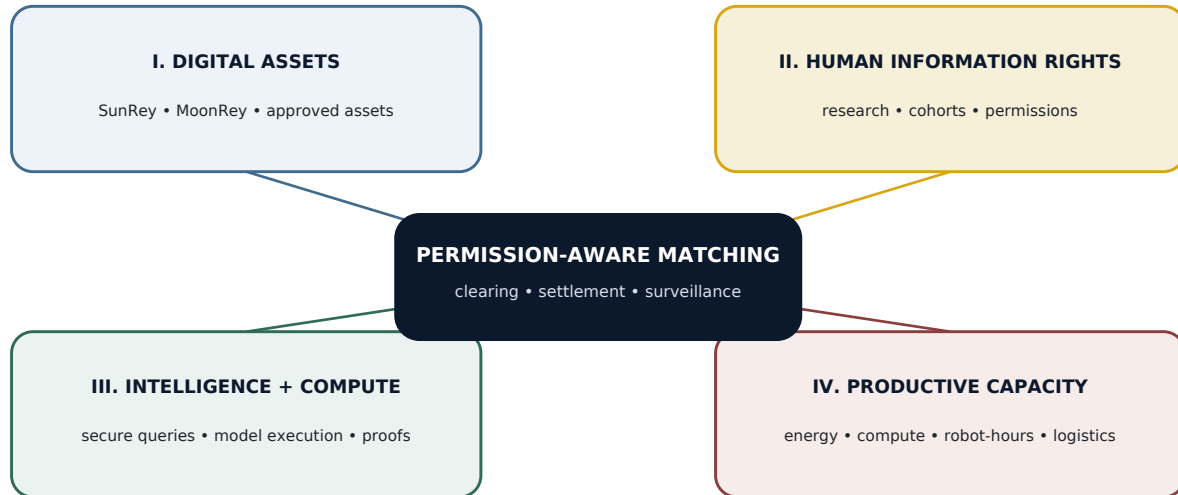


Figure 6. Four long-term market families share a common discovery, policy, clearing, settlement and evidence layer.

SunRey Exchange is designed as a Universal Economic Exchange: one market infrastructure capable of hosting multiple families of economic objects, each with its own custody, policy, pricing, delivery and evidence model. The design begins with digital-asset markets but extends to permissioned human-information rights, intelligence and compute, and MoonRey productive-capacity markets.

Conventional order books remain useful for fungible assets. Other object classes may require auctions, requests for quote, capacity reservations, sealed bids, bilateral matching, cohort requests or compute jobs. The architectural goal is not to force every market into one trading mechanic, but to give different economic objects a shared trust and settlement fabric.

EXCHANGE THESIS

One market infrastructure, multiple economic object types - each retaining its own rights, custody, policy, delivery and evidence model.

14. Four Market Families

Market family	Representative instruments / requests	Primary market form
I. Digital Assets	SunRey Coin, MoonRey Coin, approved external digital assets, permitted tokenized instruments.	order book; RFQ; auction where appropriate
II. Human Information Rights	research participation, verified cohorts, time-limited compute rights, AI-training permissions, attestations, provenance certificates.	cohort request; rights offer; bilateral / auction
III. Intelligence & Compute	secure analytics, Personal Oracle queries, approved model execution, confidential-compute proofs, inference capacity, derived intelligence.	compute job; capacity quote; service market
IV. Productive Capacity	energy, storage, GPU/CPU, robot-hours, factory time, manufacturing output, warehouse/freight, water, agriculture, real-estate and infrastructure capacity.	reservation; RFQ; auction; conditional delivery

The four-family structure is deliberately broader than a conventional crypto exchange. It creates a common discovery and settlement layer for rights and capacity without claiming that all of those objects are legally or economically equivalent.

Market I emphasizes fungible value. Market II emphasizes purpose and permission. Market III emphasizes computation, service execution and controlled outputs. Market IV emphasizes capacity, availability, performance and delivery. Their shared infrastructure is identity, policy, provenance, evidence and authoritative settlement.

OBJECT-SPECIFIC MARKETS

The Exchange can share infrastructure without erasing differences between a token, a permission, a compute job and a delivered unit of productive capacity.

15. The Core Exchange Stack

Component	Role
Market Gateway	Order, quote, capacity, auction and compute-job submission APIs.
Pre-Trade Policy Gateway	Identity, jurisdiction, eligibility, purpose, consent, sanctions/AML hooks and asset restrictions.
Order / Rights Management	Lifecycle for orders and nonstandard rights objects, expirations, reservations and cancellations.
Matching & Auction Engines	Order books plus RFQ, auction, sealed-bid, capacity and bilateral matching models.
Market Data	Prices, depth, clearing information, demand curves, capacity availability and non-sensitive indices.
Custody / Rights Registry	Segregated digital-asset custody and authoritative ownership/rights state for native objects.
Clearing & Settlement	Atomic or conditional settlement on SunRey Blockchain; delivery-versus-payment where supported.
Reconciliation	Verification between chain state, custody, external providers, oracle delivery state and accounting.
Market Surveillance	Wash trading, spoofing, self-trading, manipulation, coordinated accounts and nontraditional abuse.
Evidence Integration	Decision, order, match, policy, delivery and settlement evidence for reconstruction.

This architecture makes the Exchange a controlled market fabric rather than a standalone matching engine. Policy evaluation is pre-trade. Rights lifecycle is first-class. Settlement can be conditional on delivery. Reconciliation compares multiple authorities rather than assuming a single external provider is always correct.

MARKET CONTROL PRINCIPLE

The Exchange may discover and match economic interests, but policy, custody, rights state, settlement and evidence remain independent control responsibilities.

16. Permission-Aware Matching and Purpose-Bound Rights

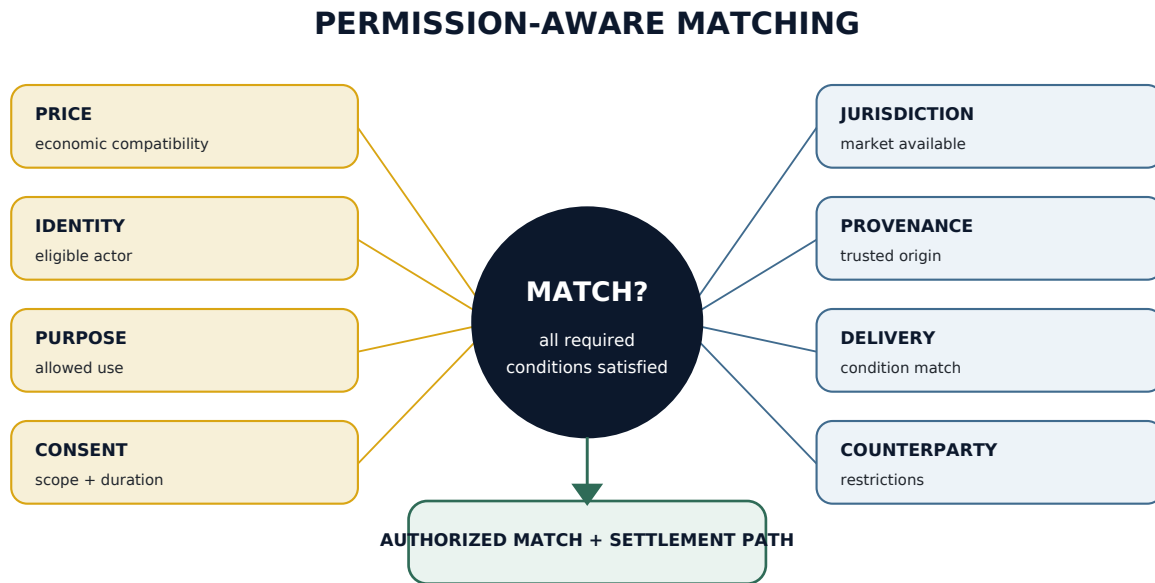


Figure 7. A valid match can require economic compatibility and permission compatibility at the same time.

Permission-aware matching is one of the most differentiated parts of the design. A trade can require more than compatible price and quantity. Matching rules can incorporate identity, jurisdiction, declared purpose, consent scope, asset eligibility, delivery conditions and counterparty restrictions before an order becomes executable.

Purpose-bound economic rights extend the same concept to information and compute. A right can authorize a defined operation for a declared purpose without becoming unrestricted ownership of the underlying data. Rights can expire by time, number of queries, quantity of capacity consumed, event completion or revocation.

WHY THIS MATTERS

Markets for sensitive information, AI computation and regulated productive assets cannot be safely modeled as “whoever holds the token can do anything.” Rights and purpose must be enforceable constraints.

17. Compute-to-Data Settlement

COMPUTE-TO-DATA: PERMISSION MOVES, COMPUTATION MOVES TO DATA

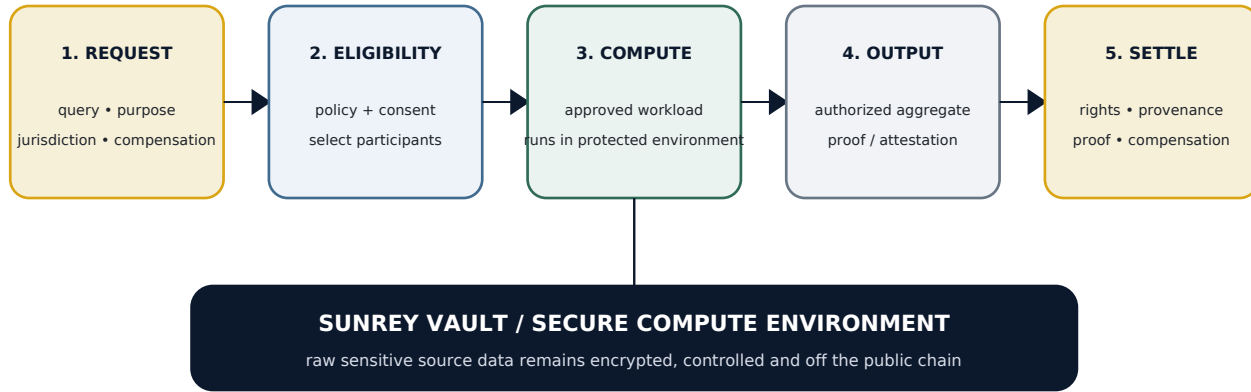


Figure 8. The buyer receives the permitted result while raw source records remain in a protected environment.

Compute-to-data inverts the conventional data-marketplace model. Instead of transferring unrestricted copies of sensitive records, an enterprise can define a narrow query, purpose, jurisdiction, compensation and output requirement. Policy and consent systems determine eligibility. Approved computation executes against protected data, and the requester receives only the permitted aggregate, proof or derived output.

SunRey Blockchain can then record permission references, provenance, proof, settlement and compensation while leaving the underlying sensitive source material off-chain. This makes the economic object a right to perform an authorized computation, not a permanent claim to the person or source record.

CRITICAL PRIVACY RULE

Tokenize rights around information, not irreversible ownership of a person's raw records. Computation should move to protected data where possible; proof and settlement move through the economic protocol.

18. Productive Capacity Discovery

MoonRey markets are designed to make productive capacity discoverable as a first-class economic object. The relevant state is not merely the theoretical value of an asset, but its current ability to provide useful output.

Capacity dimension	Illustrative state
Identity	which facility, machine, fleet or service is being referenced
Control	owner / operator / controller and recognized authority
Unit	kWh, GPU-hour, robot-hour, machine-hour, room-night, freight unit or other measurable capacity
Availability	quantity available and time window
Utilization	committed versus idle capacity
Location	physical or logical delivery location
Quality / performance	service level, reliability, latency, output tolerance or other domain measure
Provenance	source of capacity state and how it was verified
Economic terms	price, fees, compensation, settlement asset and delivery conditions
Evidence	oracle fact, attestation, receipt or proof of delivered output

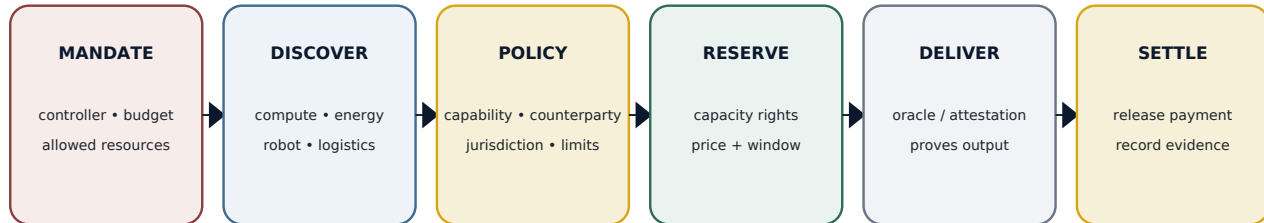
A market can then discover not just “who owns a data center” but “which verified GPU capacity is available in a permitted jurisdiction, during a required window, at a given performance level and price.” The same pattern can apply to energy, robotics, factories, logistics, water, lodging and other productive systems.

PRODUCTIVE-STATE THESIS

The protocol can represent the measurable ability to produce or deliver economic output - not only static ownership of an underlying asset.

19. Machine-Native Market Access

MACHINE-NATIVE PROCUREMENT UNDER A STRUCTURED MANDATE



THE AGENT SEARCHES. INFRASTRUCTURE-ENFORCED AUTHORITY DECIDES.

Machine commerce is bounded by controller, capability, budget, counterparty, jurisdiction and delivery conditions.

Figure 9. Machine commerce begins with bounded authority, not an unrestricted wallet.

An AI agent or machine can discover productive inputs under a structured mandate. It can compare permitted compute providers, energy sources, logistics routes or other capacity, but final execution remains constrained by controller authority, capability, budget, counterparty, jurisdiction, policy and delivery rules.

This distinction is foundational for machine-to-machine commerce. A bot that merely holds a private key has cryptographic control but no explicit economic mandate. SunRey instead models the actor's identity and authority as part of the transaction context, allowing the system to enforce what resources the agent may procure, how much it may spend and under which conditions settlement may occur.

AGENTIC ECONOMY PRINCIPLE

AI may optimize across markets, but machine agency does not eliminate governance. Capability and policy remain infrastructure-enforced constraints.

20. Multi-Resource Procurement and Conditional Settlement

A mature autonomous economy often requires several inputs at once. A single workflow may need compute, energy, information rights, robot time, manufacturing capacity and logistics. The SunRey object model allows these resources to be represented as interdependent economic inputs rather than unrelated purchases.

Stage	Illustrative autonomous manufacturing workflow
1. Design authority	An authorized design file or production specification is registered for permitted use.
2. Planning	An AI planning service constructs a resource and sequencing plan within a mandate.
3. Capacity reservation	Factory time, robotic production, compute, energy and materials are reserved as typed rights.
4. Delivery verification	Oracles and attestations verify that inputs and production milestones were actually delivered.
5. Conditional settlement	Payment or native settlement releases when the configured delivery conditions are satisfied.
6. Logistics	Freight, warehouse and final delivery capacity settle under their own conditions.
7. Provenance inheritance	The final output retains links to authorized inputs, production evidence and settlement history.

Conditional settlement is therefore not only an escrow feature. It binds economic state to verified performance. A capacity right can be reserved before delivery, while final settlement can depend on an oracle or attestation proving that the energy, compute, manufactured output or other service was actually provided.

DELIVERY-LINKED VALUE

Price describes the bargain. Delivery evidence determines whether the bargain completed. The state machine is designed to preserve both.

21. Illustrative Flow: Permissioned Human Information

The Human Information Network use case shows how economic value can be created without publishing sensitive source data.

Step	Protocol / market action
1. Request	Enterprise defines a narrow query, authorized purpose, jurisdiction, compensation and required output.
2. Eligibility	Policy and consent systems identify eligible data categories and participants.
3. Authorization	Where required, users explicitly authorize participation under defined terms.
4. Compute	Approved computation executes inside SunRey Vault or another approved secure-compute environment.
5. Output	Requester receives the authorized aggregate, proof, attestation or derived result - not unrestricted raw records.
6. Settlement	SunRey Blockchain records rights references, provenance, proof, settlement and compensation.

The protocol value is in the chain of authorization and evidence. The requester can prove the right under which computation occurred; the participant can have compensation and consent state represented; the operator can prove the policy version and settlement event; and the sensitive record can remain outside the public ledger.

CORE PRIVACY PATTERN
Data stays private. Proof moves. Permission moves. Computation moves to data where possible. Money or native value moves as settlement.

22. Illustrative Flow: Productive-Capacity Procurement

The MoonRey use case shows how an autonomous actor can purchase productive inputs without receiving unrestricted financial authority.

Step	Protocol / market action
1. Mandate	Controller specifies allowed resources, budget, risk and jurisdiction.
2. Discovery	Agent searches permitted providers for compute, energy or other required capacity.
3. Match	Exchange evaluates price plus identity, counterparty, capability, policy and delivery conditions.
4. Reserve	A Productive Capacity Right reserves the required quantity and availability window.
5. Deliver	Authenticated oracle or attestation state demonstrates that the capacity or output was supplied.
6. Settle	Payment releases under the configured condition; rights and settlement state are finalized.
7. Evidence	The event retains actor, policy, provenance, delivery and settlement references for reconstruction.

This is the foundation for machine commerce in the SunRey model. The AI system does not become sovereign because it can transact. It remains a bounded economic actor whose actions are legible to the protocol and attributable to a controller and policy context.

The same architecture can extend to multi-resource workflows in which several reservations and settlements are linked. Each leg can have its own provider, object type and evidence while the composite workflow preserves dependency and provenance.

MACHINE COMMERCE, NOT BOT CUSTODY

The objective is not to give autonomous systems unrestricted wallets. It is to make bounded machine procurement auditable, policy-aware and delivery-linked.

23. Security, Privacy, Compliance and Evidence

A protocol representing money, human-information rights and machine commerce must be designed as critical economic infrastructure.

Control domain	Required posture
Key & Validator Security	HSM/MPC support where appropriate; key rotation; validator hardening; offline recovery; privileged-access controls.
Runtime Security	Deterministic execution; resource limits; safe upgrades; independent audits; formal verification for critical modules where appropriate.
Privacy	Raw sensitive information stays off-chain or in approved secure environments; chain stores commitments, permissions, attestations, provenance and settlement evidence.
Identity Security	Passkeys/MFA for human control; strong service identity; device/agent binding; revocation and compromise recovery.
AI Agent Security	Capability tokens; no unrestricted execution credentials; per-action authorization; budgets, risk limits, model references and kill switches.
Oracle Security	Multiple-source validation; signed feeds; freshness; dispute mechanisms; anomaly detection; separation of facts from economic interpretation.
Market Integrity	Surveillance; self-trade controls; wash-trade and manipulation detection; rate limits; circuit breakers and market kill switches.
Compliance Integration	Jurisdiction packs; asset registries; transaction monitoring; sanctions/AML hooks where applicable; licensing/capability and legal-review state.
Evidence	Tamper-evident records linking intent, policy, model/agent, execution, delivery and settlement.

CRITICAL PRIVACY RULE

SunRey should tokenize rights around information, not irreversible ownership of a person's raw records. The economic object can be a permission, proof, compute right or compensation claim while the underlying sensitive record remains protected.

24. Evidence as an Economic Primitive

Evidence is not an after-the-fact audit log in the SunRey model. It is part of the architecture that explains why an economic state transition was valid. High-context markets require a durable relationship between intent, authorization, external facts, execution, delivery and settlement.

Evidence class	What it can establish
Identity / credential proof	The actor or subject satisfied a required identity, role, credential or eligibility state.
Consent / rights reference	The use was authorized under a defined scope, purpose, duration and revocation condition.
Policy reference	The exact rule or capability state used to permit or block the action.
Oracle / attestation proof	An external fact, capacity state, quality condition or delivery event was observed and authenticated.
Execution receipt	The authorized operation was submitted and executed under a known capability.
Settlement receipt	Consideration moved and final economic state was recorded.
Provenance lineage	The output can be traced to source objects, transformations, models, machines and prior rights.

The chain should not become a warehouse for every source document or telemetry stream. It should commit the minimum cryptographic state required to reconstruct the authorization and economic meaning of the event, while sensitive or high-volume evidence remains in protected stores.

SETTLEMENT GRAPH, NOT DATA WAREHOUSE

Keep private documents, full health records and industrial telemetry in specialized protected systems.
Keep the cryptographic commitments required to prove identity, rights, policy, provenance, ownership, delivery and settlement on the sovereign economic layer.

25. Protocol Governance and Interoperability

GOVERNANCE MUST BE SEPARATED BY AUTHORITY DOMAIN

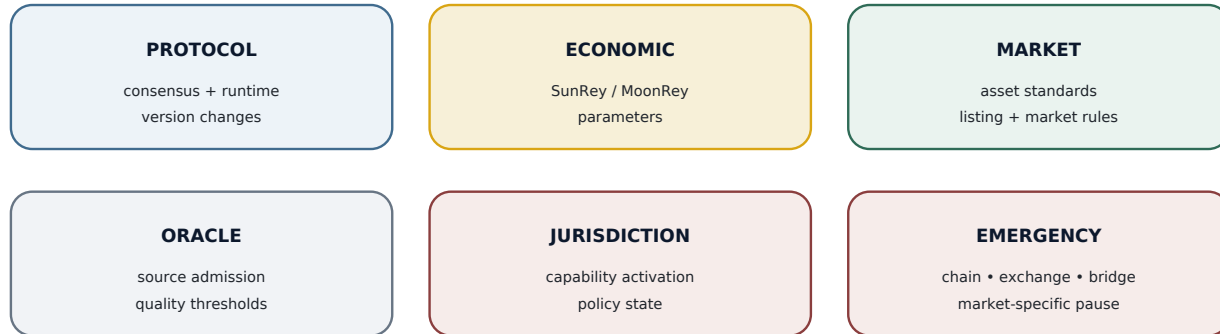


Figure 10. Technical, economic, market, oracle, jurisdiction and emergency authorities should not collapse into one governance mechanism.

Protocol governance must distinguish consensus/runtime upgrades, native-asset economic parameters, validator policy, asset standards and exchange listings, oracle-source quality, jurisdictional capability and emergency response. Critical upgrades should use staged activation, code review, test networks, delay windows and rollback or pause strategies where feasible.

SunRey is intended to be sovereign without being isolated. External blockchains, banking, card, payment, cloud, data, energy and infrastructure systems can connect through explicit adapters and gateways. Bridges should be minimized, monitored and treated as high-risk infrastructure. External networks are integration targets, not the authoritative base layer for SunRey-native state.

SOVEREIGN, NOT CLOSED

Interoperability expands utility. It should not transfer root authority over SunRey-native identity, rights, assets or settlement to an external network.

26. Strategic Differentiation

The architecture is deliberately not positioned around generic blockchain metrics such as more token types, lower gas or headline throughput. Those characteristics are replicable. The differentiated thesis is the integrated economic state model.

Conventional category	Typical focus	SunRey differentiation
Generic smart-contract chain	Programmable state and token applications.	Identity, permission, productive capacity, AI-agent capability, policy and provenance become first-class economic concepts.
Crypto exchange	Spot/derivatives trading of fungible assets.	Exchange extends to rights, compute, human-information participation and productive-capacity markets.
Data marketplace	Copies or licenses datasets.	Compute-to-data, purpose-bound rights, attestations and participant compensation without unrestricted raw-data transfer.
RWA tokenization	Ownership or financial claims on real-world assets.	MoonRey can additionally represent verified productive capacity and delivered output, not only static ownership.
AI-agent wallet	AI controls a wallet or account.	Machine identity, capabilities, budgets, risk, policy and evidence are infrastructure-enforced.
Fintech app	Aggregates banking, payments, investing and data.	Financial operating system can connect to a sovereign human-information and autonomous-production protocol.

DEFENSIBILITY THESIS

The potential moat is the economic ontology and verification framework: the machine-readable relationship between actors, rights, purpose, policy, provenance, productive state, settlement and evidence.

27. Protocol Maturity Path

The source architecture uses a simulation-first roadmap that separates protocol experimentation from live regulated or economically irreversible activity.

Phase	Primary deliverable
0 - Protocol Specification	Canonical objects; actor identity; SunRey/MoonRey rules; state model; transaction envelope; consensus/runtime ADRs; threat model.
1 - Local Sovereign Chain	Single-node and local multi-validator simulation; deterministic runtime; native asset ledger; identity registry; event/evidence model.
2 - Permission & Provenance	Rights registry; consent; attestations; provenance; policy hooks; agent capabilities; secure fixtures.
3 - Exchange Simulation	Order management; matching; market data; surveillance; clearing and reconciliation using simulated assets.
4 - Human Information Market Simulation	Compute-to-data jobs; consented research; proof of contribution; synthetic users/data.
5 - MoonRey Capacity Market Simulation	Synthetic energy, GPU, robot and manufacturing capacity; oracles; reservations; delivery-linked settlement.
6 - Public / Partner Testnet	Independent validators; security audits; load/failure tests; bridge prototypes; oracle pilots; economic attack simulations.
7 - Regulated Pilot Markets	Only capabilities with identified legal entity, partner/license basis, audited custody, compliance controls and approved policies.
8 - Multi-Market Production	Progressive activation by jurisdiction, market type and risk class.

BUILD RULE

Do not activate live custody, exchange trading, human-information monetization or real-world asset claims because the software exists. Production capability remains default-deny until legal, security, operational and partner requirements are satisfied.

28. Open Architecture Decisions

The economic thesis is sufficiently defined to guide protocol specification, but several foundational technology and governance choices remain intentionally open.

Decision	Questions still to resolve
Consensus	Validator-set model; permissionless vs permissioned/hybrid path; finality; accountability; geographic distribution.
Execution Runtime	WASM or other deterministic runtime; languages; metering; upgrades; native modules vs general contracts.
State & Storage	Account vs object model; Merkle/state structure; archival strategy; pruning; indexing; evidence references.
Privacy	Commitments; verifiable credentials; zero-knowledge proofs; confidential compute; selective disclosure.
Identity	Credential standards; recovery; pseudonymity; machine identity; legal-identity binding; privacy boundaries.
Oracle Architecture	Provider admission; signatures; aggregation; disputes; freshness; asset-specific verification.
SunRey / MoonRey Economics	Supply; issuance; fees; staking/security role; treasury; rewards; governance; sinks; asset segregation.
Exchange Market Models	Order book vs RFQ/auction/reservation by object type; clearing; delivery; market-maker roles.
Interoperability	External chain bridges; fiat/payment adapters; messaging standards; failure isolation.
Governance	Upgrade authority; validator/community roles; emergency councils; parameter governance; legal-entity separation.
Performance	Finality targets; throughput; data availability; cost model; high-frequency machine commerce.
Regulatory Topology	Sovereign Cells; regional validators; data residency; asset restrictions; market activation by jurisdiction.

Next formal technical deliverable: a SunRey Blockchain Technical Architecture Specification covering the formal state model, transaction schema, object standards, consensus and execution ADRs, native-asset economics, identity, oracle architecture, Exchange protocols, threat model and phased testnet plan.

29. Conclusion

SunRey Blockchain is designed as sovereign economic infrastructure for a dual human-and-machine economy. SunRey Coin coordinates the human-information and participation layer. MoonRey Coin coordinates autonomous productive capacity and machine-generated output. SunRey Exchange connects these domains through markets for digital assets, permissioned rights, intelligence, compute and productive capacity.

The central architectural decision is to make economic context enforceable rather than descriptive. Identity, permission, purpose, provenance, jurisdiction, capability, delivery and evidence can be represented as machine-readable state that execution infrastructure evaluates before settlement and can reconstruct afterward.

Sensitive information remains protected; proofs and rights can be portable. AI agents and robots can participate economically, but only inside explicit capabilities and policy. Productive assets can become digitally discoverable and contractible without assuming that every token is a legal claim on a physical asset.

SUNREY THESIS

A sovereign blockchain for a post-AI economy in which humans, AI agents, machines, information rights, energy, compute, resources and productive capacity can be represented, permissioned, exchanged and settled within one coherent economic protocol.

If executed successfully, the architecture is not merely another blockchain or another exchange. It is intended to become an operating and settlement layer for the relationship between human economic value and autonomous productive value.

A. Terminology and Source Basis

Terminology used in this paper follows the current SunRey naming and architecture direction.

Term	Meaning
SunRey Blockchain	Proprietary sovereign blockchain and Web3/economic base layer for the ecosystem.
SunRey Coin	Native asset coordinating the human-information and participation economy.
MoonRey Coin	Native asset coordinating the autonomous productive-capacity economy.
SunRey Exchange	Universal economic market layer for digital assets, human-information rights, intelligence/compute and productive capacity.
Human Information Network	Permission-aware information and intelligence network centered on individual control, verification, compute-to-data and economic participation.
Global Productive Capacity Graph	Machine-readable model of energy, compute, robots, factories, resources, infrastructure and productive capacity.
PEVE	Personal Economic Value Engine; market-oriented valuation of specific authorized contributions, not a score of human worth.
Economic State Machine	Protocol model in which actors, rights, purpose, policy, provenance, productive state, settlement and evidence are first-class economic context.

Primary architecture source

SunRey Blockchain & Universal Economic Exchange - Technical Architecture and Novel Market Design, Version 1.0, August 2026. This document supplies the core state-machine thesis, protocol component model, native economic object taxonomy, Exchange architecture, differentiated market mechanisms, illustrative flows, security/evidence model, governance, interoperability, roadmap and open architecture decisions.

Supporting architecture sources

SunRey + MoonRey - Dual-Economy Architecture for a Post-AI, Post-Automation World, August 2026; SunRey Comprehensive Business Plan, September 2026, used only for current public naming and concise description of the Sovereign Economic State Machine; SunRey Strategy Report, August 2026, for the Human Information Network and privacy-preserving rights model.

PUBLICATION NOTE

This white paper presents conceptual protocol architecture and design intent. It does not assert that unresolved consensus, economics, privacy, oracle, governance or regulatory decisions are finalized, and it does not create legal rights in any asset or productive resource.